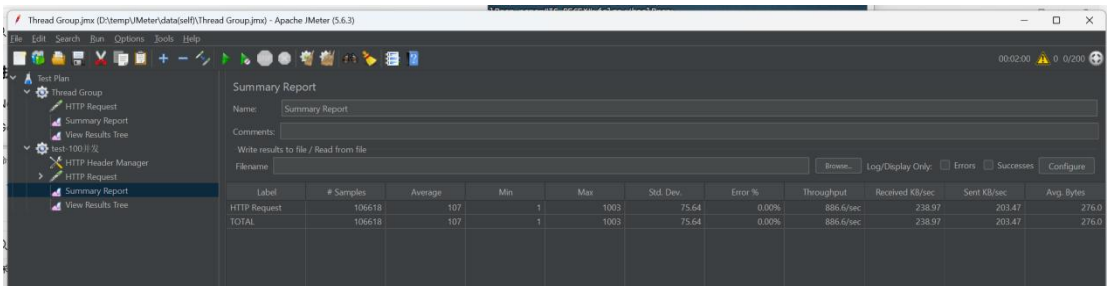
三结点部署正常

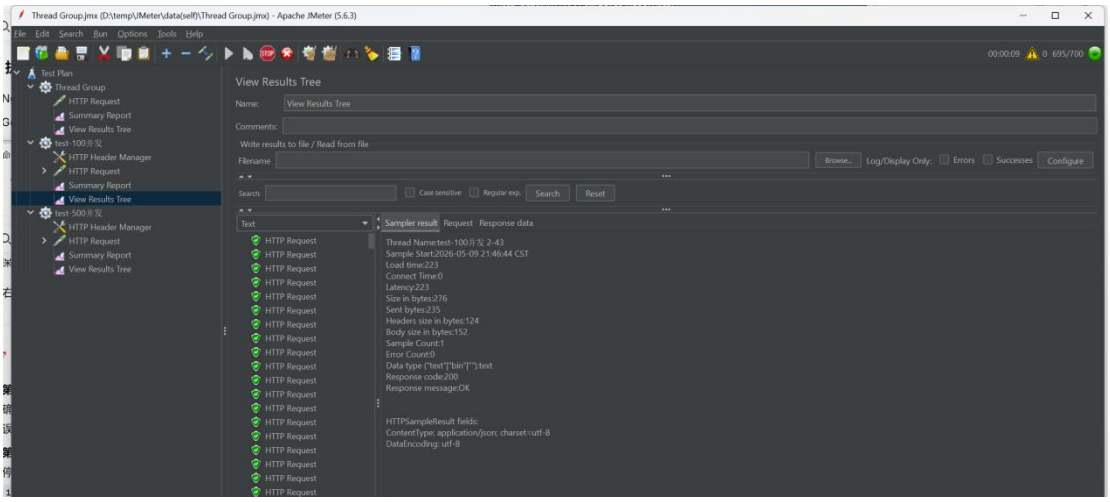
```
[root@k3s-node1 ~]# ln -sf /usr/local/bin/k3s /usr/local/bin/kubectl
[root@k3s-node1 ~]# kubectl get nodes
NAME        STATUS   ROLES                  AGE     VERSION
k3s-node1   Ready    control-plane,master   6m12s   v1.27.16+k3s1
k3s-node2   Ready    <none>                 2m28s   v1.27.16+k3s1
k3s-node3   Ready    <none>                 109s    v1.27.16+k3s1
[root@k3s-node1 ~]# kubectl get nodes -o wide
NAME        STATUS   ROLES                  AGE     VERSION         INTERNAL-IP      EXTERNAL
-IP   OS-IMAGE               KERNEL-VERSION          CONTAINER-RUNTIME
k3s-node1   Ready    control-plane,master   7m41s   v1.27.16+k3s1   192.168.64.141   <none>
      CentOS Linux 7 (Core)   3.10.0-957.el7.x86_64   containerd://1.7.17-k3s2.27
k3s-node2   Ready    <none>                 3m57s   v1.27.16+k3s1   192.168.64.142   <none>
      CentOS Linux 7 (Core)   3.10.0-957.el7.x86_64   containerd://1.7.17-k3s2.27
k3s-node3   Ready    <none>                 3m18s   v1.27.16+k3s1   192.168.64.143   <none>
      CentOS Linux 7 (Core)   3.10.0-957.el7.x86_64   containerd://1.7.17-k3s2.27
[root@k3s-node1 ~]#
```

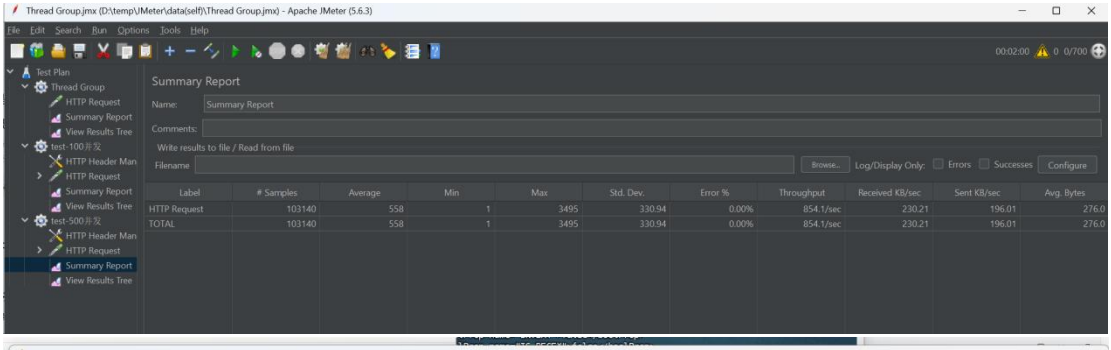
各服务正常运行

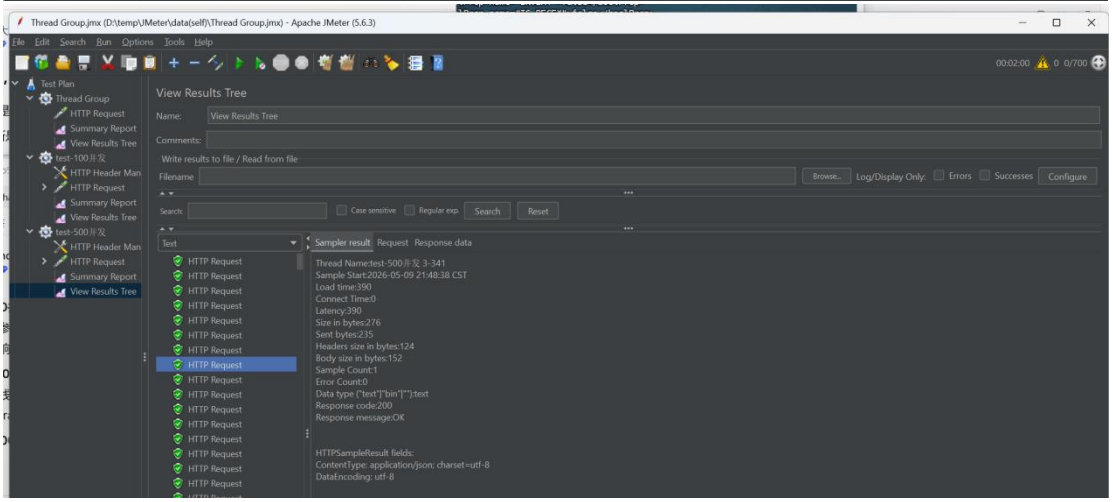
```
.sh >> /var/log/hdfs_backup.log 2>&1") | crontab -
[root@k3s-node1 ~]# kubectl get pods -n schedule-system
NAME                                  READY   STATUS    RESTARTS       AGE
alertmanager-5cbf8fcd5d-9h52p         1/1     Running   1 (133m ago)   9h
go-schedule-service-786fdbf9fc-nlwh5  1/1     Running   2 (133m ago)   6d1h
grafana-885f5c6cf-d6mws               1/1     Running   2 (133m ago)   6d
kafka-5c588768fd-tdpqm                1/1     Running   2 (133m ago)   10h
mysql-0                               1/1     Running   2 (133m ago)   6d5h
mysql-1                               1/1     Running   2 (133m ago)   6d5h
mysql-2                               1/1     Running   2 (133m ago)   6d5h
nginx-db8df4f54-nlf8g                 1/1     Running   4 (133m ago)   6d
nginx-db8df4f54-qcw4k                 1/1     Running   4 (133m ago)   6d
prometheus-76d76dd497-52z8m           1/1     Running   2 (133m ago)   6d
redis-0                               2/2     Running   4 (133m ago)   6d2h
redis-1                               2/2     Running   4 (133m ago)   6d2h
redis-2                               2/2     Running   4 (133m ago)   6d2h
zookeeper-c759474d-f6nnl              1/1     Running   2 (133m ago)   6d
[root@k3s-node1 ~]#
```

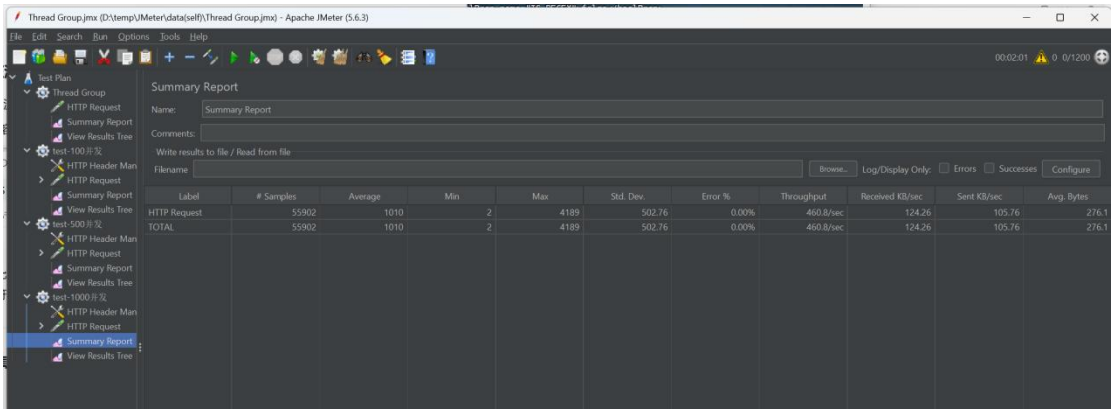
并发测试

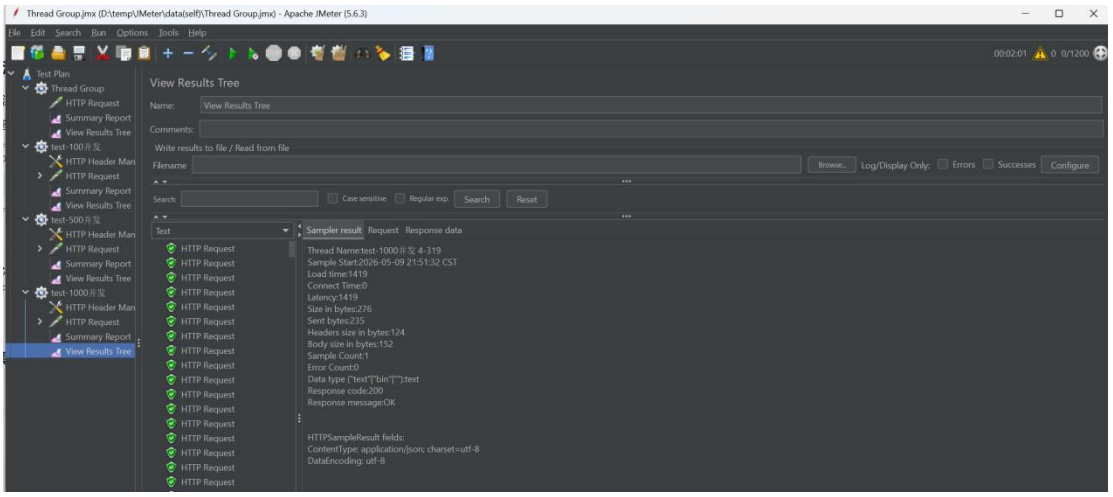






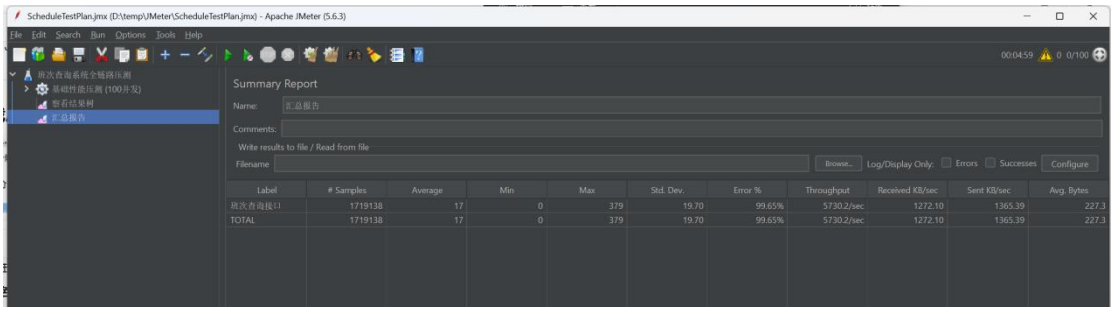


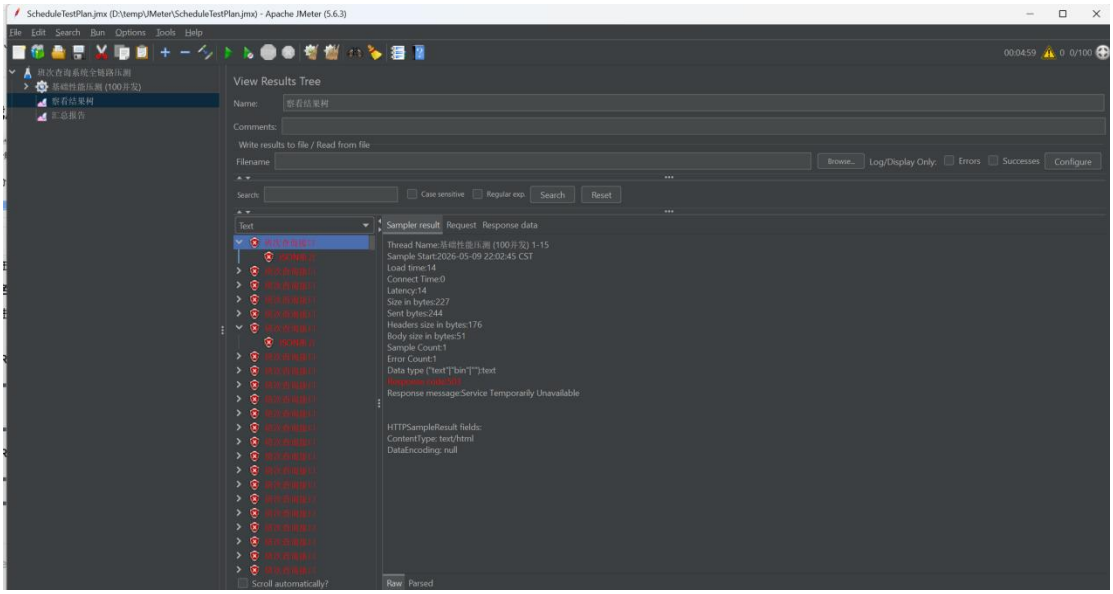




Nginx 限流保护机制工作正常，成功将过载请求拒之门外，保护了后端 Go 服

务不被瞬间流量冲垮



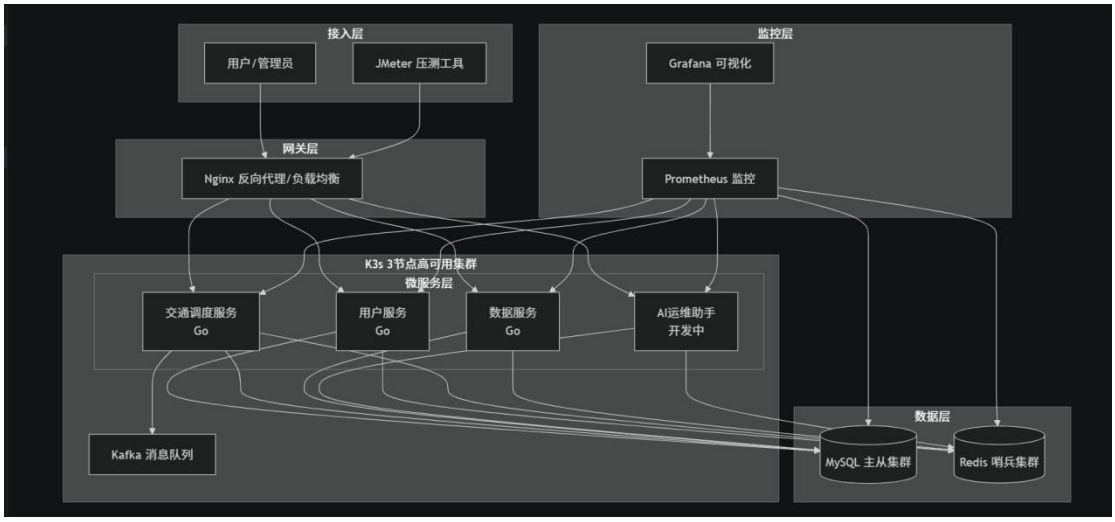


grafana 仪表盘




在线节点数
3
运行中 Pod
17
运行中容器
39
采集目标存活
4
每节点 kubelet CPU 使用（核）
每节点 kubelet 内存
每节点运行 Pod 数
每节点运行容器数
k3s-node1
k3s-node2
k3s-node3

架构图


接入层
用户/管理员
JMeter 压测工具
监控层
Grafana 可视化
网关层
Nginx 反向代理/负载均衡
Prometheus 监控
K3s 3节点高可用集群
微服务层
交通调度服务
Go
用户服务
Go
数据服务
Go
AI运维助手
开发中
数据层
Kafka 消息队列
MySQL 主从集群
Redis 哨兵集群